

# VK KAPOOR

Cloud & Applied AI Solutions Architect | AWS Certified Solutions Architect - Professional

Parker, CO | [vk.kapoor@icloud.com](mailto:vk.kapoor@icloud.com) | [linkedin.com/in/vkapoor9](https://linkedin.com/in/vkapoor9) | [vkapoor9.github.io](https://vkapoor9.github.io)

## PROFESSIONAL SUMMARY

Solutions Architect and hands-on AI builder with 20+ years specializing in large-scale cloud migrations. As an AWS-certified architect, I lead complex on-premises-to-AWS migrations for government and enterprise, designing to the Well-Architected Framework: I have moved 55+ applications across 2,500+ EC2 instances and cut cloud spend by \$760K a year. Over the past year I have brought that same end-to-end discipline to AI, building 30+ applications with Claude Code, including two live SaaS products and eight specialized AI agents that run on both local and frontier models. Across all of it, I lead the teams and stakeholders that take ambitious technical bets from architecture to working, delivered software.

## KEY SKILLS

**Cloud Architecture & Migration:** AWS (EC2, S3, RDS, Aurora, VPC, Lambda, Direct Connect, CloudFront, ALB/NLB, Auto Scaling, Organizations, Control Tower), Azure, Multi-Cloud, Hybrid Cloud, GovCloud, AWS Well-Architected Framework

**AI & GenAI:** AI Application Architecture, Prompt Engineering, LLM Integration & Tool Use, Autonomous AI Agents & Multi-Agent Orchestration (BMAD), AI-Augmented Development (Claude Code, GitHub Copilot), Model Selection across Providers (Anthropic Claude, OpenAI / GPT, local models via Ollama), Privacy-First / On-Device Inference, Voice-Enabled Apps; exploring Amazon Bedrock & Amazon Q

**Security & Compliance:** IAM, Security Hub, WAF, GuardDuty, AWS Config, Secrets Manager, Systems Manager, DISA STIG, SCAP, Zero Trust Architecture, FedRAMP, continuous ATO

**DevOps, IaC & Containers:** Terraform, CloudFormation, Ansible (architecture & review), Git, CI/CD pipeline design, Docker, Proxmox VE, VMware; Solution Design: Kubernetes, EKS, ECS, Fargate

**FinOps & Tooling:** Cost Explorer, Trusted Advisor, Budgets, Savings Plans, CloudWatch, CloudTrail; Python and SQL (AI-assisted), YAML, JSON, Jira, Confluence, ServiceNow

## PROFESSIONAL EXPERIENCE

### Engineering Manager II | Deloitte Consulting LLP | Denver, CO

Jan 2019 - Present

#### Lead Cloud Architect - State Driver-Services System Migration (State Government), 2025 - Present

- Leading a cross-functional team of 15 (engineers, architects, developers, DBAs, and analysts) migrating a mission-critical state driver-services system serving a population of ~31 million residents to AWS; designing end-to-end Well-Architected infrastructure across 10-15 AWS accounts (EC2, RDS, S3, VPC, IAM, CloudWatch) with AI-driven project management and application modernization.
- Defining DevSecOps requirements including STIG scanning and Security Hub integration across all environments; partnering with agency stakeholders from executives to engineers, and using Claude Code, GitHub Copilot, and the BMAD multi-agent framework in daily development to accelerate delivery.

#### Deputy Cloud Architect - State Health & Human Services Benefits System Migration, 2023 - 2024

- Designed high-availability infrastructure (EC2, RDS, VPC, Transit Gateway, ALB/NLB, Auto Scaling) on the AWS Well-Architected Framework for a benefits-eligibility platform serving millions of residents across Medicaid and SNAP; drove FinOps optimization achieving ~50% cost reduction (\$760K/yr) via right-sizing, Savings Plans, and Cost Explorer.
- Led the unstructured data migration workstream with hands-on NetApp ONTAP CLI administration (SnapMirror) into AWS FSx for NetApp ONTAP and EFS.

#### Cloud Solutions Architect - Federal Housing-Finance Agency, FedRAMP-Authorized GovCloud, 2021 - 2023

- Architected compute, storage, networking, and migration for a federal housing-finance system supporting millions of U.S. households; executed a Disaster Recovery POC with AWS CloudEndure across GovCloud regions and automated SCAP/STIG compliance, maintaining continuous ATO readiness across four environments.

### Sr. Technical Consultant / Manager | Larsen & Toubro Infotech | Denver, CO

Oct 2014 - Jan 2019

- Initiated and delivered a custom mobile app for asset/work management, saving the client \$300K/year in licensing costs; supervised development teams across requirements, design, QA, and deployment.

- Executed a full-stack enterprise platform upgrade including database migration (SQL Server 2008 to 2016), IBM WebSphere upgrade, and horizontal/vertical cluster scaling.

## Earlier Experience | IBM, MRO Software, and others | Bedford, MA & Denver, CO

2003 - 2014

- Progressive engineering and technical support roles spanning enterprise asset management, application servers (WebSphere, WebLogic), databases (Oracle, SQL Server, DB2), and systems administration.

## AI PROJECTS & BUILDS

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- **Public portfolio ([vkapoor9.github.io](https://vkapoor9.github.io)):** 30+ AI tools and applications built end to end with Claude Code, including 2 live SaaS products and 8 specialized AI agents, every project shipped and used. Selected builds below; full detail and live demos on the site.
- **Privacy-first family AI agent:** 8 agent roles and 23+ voice tools with 60+ tests, routing per task between local and hosted models based on a privacy classification, so sensitive data stays on-device.
- **Kid-safe AI tutor:** used daily by 7 children, with fully on-device inference and speech so kids' data never leaves the home network.
- **Live AI content SaaS:** real customers, OAuth and payments in production, generating 22 platform-tuned content variations across 11 channels from a single input.
- **On-device macOS voice assistant:** a fully local assistant on Apple Silicon with sub-second response and 383 passing tests, plus AI note-taking, automation dashboards, and browser games.

## CERTIFICATIONS & EDUCATION

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[AWS Solutions Architect - Professional](#) | [AWS Security - Specialty](#) | [Microsoft Azure Fundamentals \(AZ-900\)](#) | PMP (certifications verifiable on Credly)

MBA - Karnataka State Open University, India, 2011 | B.Sc. Computer Science - UMass Lowell, USA, 2005

## LANGUAGES & INTERESTS

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Languages: English, Hindi, Punjabi (fluent); Japanese (learning).

Interests: building AI side projects for the fun of it, AI-assisted music and photography, home automation and homelab, and the Apple ecosystem.